

Airbus Product Standard

Carbon fibre reinforced epoxy prepreg Plain weave fabric / 180°C – curing class Flow Controlled

Structural material

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1 Scope

This standard specifies the required characteristics of EP-prepreg made from carbon fibre fabric, plain weave, preimpregnated with modified epoxy resin, flow controlled, curing at 180°C, for use in aerospace applications for manufacture of structural components.

2 Normative References

This Airbus Standard incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any these publications apply to this Airbus Standard only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

AIMS05-01-000 Part 1	Technical Specification - Carbon fibre reinforced epoxy prepreg; Part 1: General requirements.
AIMS05-01-000 Part 6	Technical Specification - Carbon fibre reinforced epoxy prepreg; Qualification programme / batch release testing – Fabric / 180°C curing class; Flow controlled; Structural material
AIMS05-01-005	Material Specification - Carbon fibre reinforced epoxy prepreg; Plain weave fabric / 180°C curing class; Flow controlled

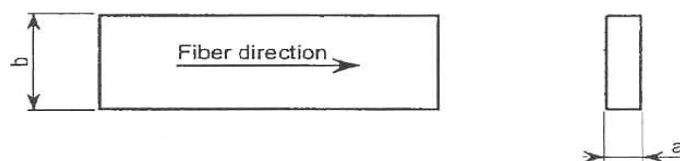
3 Requirements

3.1 Configuration, Dimensions, Tolerances and Mass

The configuration, dimensions, tolerances and mass shall be in accordance with Figure 1 and Table 1 and Table 2.

3.2 Material

See Table 1.



a= Thickness, see Table 1.

b= Width, see Table 2.

Figure 1: Configuration

Table 1: Material and Mass

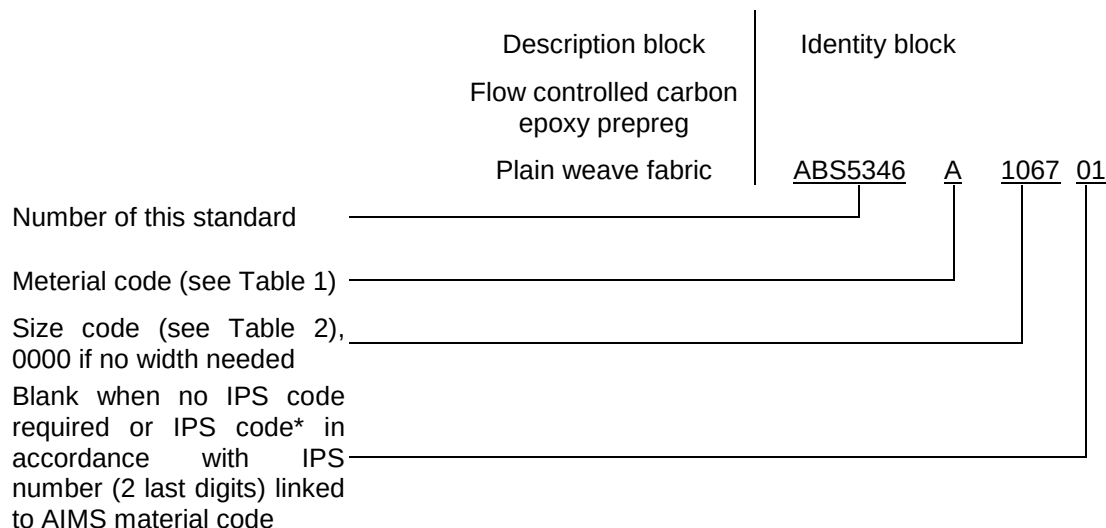
Material Code	Material Specification	Thickness per ply ¹⁾ (mm)	Resin Content (% by weight)	Prepreg areal weight (g/m ²)	Weave Style
A	AIMS05-01-005 A	0,237 ¹⁾	44 ± 3	345 ± 25	Plain weave 1/1
¹⁾ Nominal cured ply thickness for 46% FVC in accordance with the nominal resin content, 193 g/m ² fiber areal weight.					

Table 2: Dimensions and tolerances

Size code	Width
	(mm)
0000 ¹⁾	-
1067	1067 ± 10 mm
1520	1520 ± 10 mm
¹⁾ Size Code 0000 means that the width is not defined, any width is permissible. If IPS number is not specified, it is allowed not to write this code.	

4 Designation

This type of standard shall be designated according to the philosophy of the following example:



Note: * Designation with IPS code when required by design

5 Marking

Marking shall be in accordance with AIMS05-01-000 Part 1.

6 Technical Specification

AIMS05-01-000 Part 1 and Part 6.

Record of Revisions

Issue	Clause modified	Description of Modification
1 04.2005	1	New Standard
2 09.2018	All 3.2	Updated to latest template and reformatted as required Table 1: Resin content tolerance harmonized with AIMS05-01-005 Table 2: New size code 1520 added
3 11.2020	3.2	Table 1: Fiber areal weight corrected to 193 g/m ² , FVC information added to footnote